

cppulse

C++ Codebase Health Report

/tmp/poco

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1. Executive Summary

55.2Overall Health Score
Poor — significant technical debt requires attention.

200

TOTAL FINDINGS

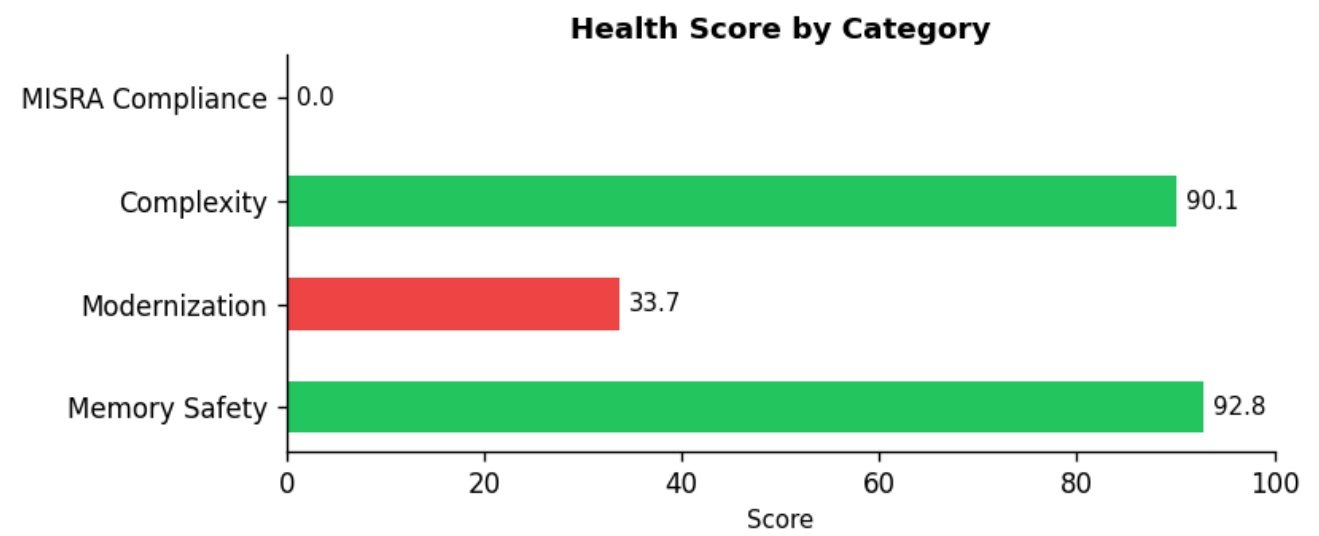
545

ERRORS

16929

WARNINGS

2. Health Score Breakdown



Category	Score (/ 100)	Status
Memory Safety	92.8	GOOD
Modernization	33.7	POOR
Complexity	90.1	GOOD
MISRA Compliance	0.0	POOR

3. Hotspot Map — Top 20 Files

Hotspot score = change frequency × complexity × debt density. High-score files deliver the greatest ROI when refactored.

#	File	Hotspot Score	Change Freq.	Findings
1	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	0.00	0	7
2	ActiveRecord/ Compiler/src/ Compiler.cpp	0.00	0	5
3	ActiveRecord/ Compiler/src/ HeaderGenerator.cpp	0.00	0	6
4	ActiveRecord/ Compiler/src/ ImplGenerator.cpp	0.00	0	11
5	ActiveRecord/ Compiler/src/ Parser.cpp	0.00	0	5
6	ActiveRecord/ include/Poco/ ActiveRecord/ Query.h	0.00	0	5
7	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	0.00	0	29
8	ActiveRecord/ testsuite/src/ ActiveRecordTestSuite.cpp	0.00	0	1
9	ActiveRecord/ testsuite/src/ WinDriver.cpp	0.00	0	1
10	ApacheConnector/ include/ ApacheApplication.h	0.00	0	3
11	ApacheConnector/ include/ ApacheConnector.h	0.00	0	1
12	ApacheConnector/ include/ ApacheRequestHandlerFactory.h	0.00	0	3

#	File	Hotspot Score	Change Freq.	Findings
13	ApacheConnector/ include/ ApacheServerReques t.h	0.00	0	2
14	ApacheConnector/ include/ ApacheServerRespon se.h	0.00	0	2
15	ApacheConnector/ include/ ApacheStream.h	0.00	0	2
16	ApacheConnector/ samples/ FormServer/src/ FormServer.cpp	0.00	0	9
17	ApacheConnector/ samples/ TimeServer/src/ TimeServer.cpp	0.00	0	3
18	ApacheConnector/ src/ ApacheConnector.cp p	0.00	0	6
19	Benchmark/src/ BenchmarkApp.cpp	0.00	0	18
20	Benchmark/src/ LoggerBench.cpp	0.00	0	21

4. Detection Findings

Modernization (9 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-004	INFO	String literal contains 3 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ActiveRecord/ Compiler/src/ CodeGenerator.cpp, ActiveRecord/ Compiler/src/ HeaderGenerator.cpp, ActiveRecord/ Compiler/src/ ImplGenerator.cpp +1 more	14
CPP-MOD-008	WARNING	Unscoped enum 'LogLevel' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum LogLevel' with 'enum class LogLevel'</i>	ApacheConnector/ include/ ApacheApplication.h, ApacheConnector/ include/ ApacheConnector.h, ApacheConnector/ include/ ApacheRequestHandlerFactory.h +3 more	6
CPP-MOD-008	WARNING	Unscoped enum '(unnamed enum at /tmp/poco/ApacheConnector/include/ApacheStream.h:36:2)' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum (unnamed enum at /tmp/poco/ApacheConnector/include/ApacheStream.h:36:2)' with 'enum class (unnamed enum at /tmp/poco/ApacheConnector/include/ApacheStream.h:36:2)'</i>	ApacheConnector/ include/ ApacheApplication.h, ApacheConnector/ include/ ApacheRequestHandlerFactory.h, ApacheConnector/ include/ ApacheServerRequest.h +2 more	5

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-006	INFO	Index-based for loop may be replaceable with a range-based for loop <i>Consider 'for (const auto& elem : container)' instead</i>	Benchmark/src/LoggerBench.cpp	3
CPP-MOD-009	INFO	typedef 'RequestHandlerFactories' should use a C++11 type alias instead <i>Replace 'typedef int RequestHandlerFactories' with 'using RequestHandlerFactories = int'</i>	ApacheConnector/include/ApacheApplication.h, ApacheConnector/include/ApacheRequestHandlerFactory.h	2
CPP-MOD-004	INFO	String literal contains 5 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ActiveRecord/Compiler/src/Compiler.cpp	1
CPP-MOD-004	INFO	String literal contains 65 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ApacheConnector/samples/FormServer/src/FormServer.cpp	1
CPP-MOD-004	INFO	String literal contains 4 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ApacheConnector/samples/TimeServer/src/TimeServer.cpp	1
CPP-MOD-004	INFO	String literal contains 8 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	Benchmark/src/BenchmarkApp.cpp	1

Misra (30 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'result' of type 'int' declared without an initializer <i>Initialize 'result' at the point of declaration</i>	ActiveRecord/ Compiler/src/ CodeGenerator.cpp, ActiveRecord/ include/Poco/ ActiveRecord/ Query.h, Benchmark/src/ PatternFormatterBenchmark.cpp	12
MISRA-007	WARNING	Variable 'msg' of type 'int' declared without an initializer <i>Initialize 'msg' at the point of declaration</i>	Benchmark/src/ LoggerBenchmark.cpp, Benchmark/src/ NotificationQueueBenchmark.cpp, Benchmark/src/ PatternFormatterBenchmark.cpp	12
MISRA-007	WARNING	Variable 'pContext' of type 'int' declared without an initializer <i>Initialize 'pContext' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	11
MISRA-007	WARNING	Variable 'formatter' of type 'int' declared without an initializer <i>Initialize 'formatter' at the point of declaration</i>	Benchmark/src/ PatternFormatterBenchmark.cpp	10
MISRA-007	WARNING	Variable 'parser' of type 'int' declared without an initializer <i>Initialize 'parser' at the point of declaration</i>	ActiveRecord/ Compiler/src/ Parser.cpp, CppParser/ testsuite/src/ AttributesParserTest.cpp	8
MISRA-007	WARNING	Variable 'attrs' of type 'int' declared without an initializer <i>Initialize 'attrs' at the point of declaration</i>	CppParser/ testsuite/src/ AttributesParserTest.cpp	7
MISRA-007	WARNING	Variable 'result' of type 'auto' declared without an initializer <i>Initialize 'result' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	6

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'source' of type 'int' declared without an initializer <i>Initialize 'source' at the point of declaration</i>	Benchmark/src/LoggerBench.cpp	5
MISRA-007	WARNING	Variable 'pSuite' of type 'int *' declared without an initializer <i>Initialize 'pSuite' at the point of declaration</i>	ActiveRecord/testsuite/src/ActiveRecordTest.cpp, ActiveRecord/testsuite/src/ActiveRecordTestSuite.cpp, CppParser/testsuite/src/AttributesParserTest.cpp +1 more	4
MISRA-007	WARNING	Variable 'text' of type 'int' declared without an initializer <i>Initialize 'text' at the point of declaration</i>	Benchmark/src/LoggerBench.cpp	4
MISRA-007	WARNING	Variable 'queue' of type 'int' declared without an initializer <i>Initialize 'queue' at the point of declaration</i>	Benchmark/src/NotificationQueueBench.cpp	4
MISRA-007	WARNING	Variable 'pRole' of type 'int' declared without an initializer <i>Initialize 'pRole' at the point of declaration</i>	ActiveRecord/testsuite/src/ActiveRecordTest.cpp	3
MISRA-007	WARNING	Variable 'message' of type 'int' declared without an initializer <i>Initialize 'message' at the point of declaration</i>	Benchmark/src/LoggerBench.cpp	3
MISRA-007	WARNING	Variable 'pNf' of type 'int' declared without an initializer <i>Initialize 'pNf' at the point of declaration</i>	Benchmark/src/NotificationQueueBench.cpp	3
MISRA-007	WARNING	Variable 'consumerThread' of type 'int' declared without an initializer <i>Initialize 'consumerThread' at the point of declaration</i>	Benchmark/src/NotificationQueueBench.cpp	3

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'type' of type 'std::string' declared without an initializer <i>Initialize 'type' at the point of declaration</i>	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	2
MISRA-007	WARNING	Variable 'self' of type 'int' declared without an initializer <i>Initialize 'self' at the point of declaration</i>	ActiveRecord/ Compiler/src/ Compiler.cpp, Benchmark/src/ BenchmarkApp.cpp	2
MISRA-007	WARNING	Variable 'options' of type 'int' declared without an initializer <i>Initialize 'options' at the point of declaration</i>	ActiveRecord/ Compiler/src/ Compiler.cpp, Benchmark/src/ BenchmarkApp.cpp	2
MISRA-007	WARNING	Variable 'helpFormatter' of type 'int' declared without an initializer <i>Initialize 'helpFormatter' at the point of declaration</i>	ActiveRecord/ Compiler/src/ Compiler.cpp, Benchmark/src/ BenchmarkApp.cpp	2
MISRA-007	WARNING	Variable 'guard' of type 'std::string' declared without an initializer <i>Initialize 'guard' at the point of declaration</i>	ActiveRecord/ Compiler/src/ HeaderGenerator.cpp	2
MISRA-007	WARNING	Variable 'it' of type 'auto' declared without an initializer <i>Initialize 'it' at the point of declaration</i>	ActiveRecord/ Compiler/src/ HeaderGenerator.cpp, ActiveRecord/ Compiler/src/ ImplGenerator.cpp	2
MISRA-007	WARNING	Variable 'result' of type 'std::string' declared without an initializer <i>Initialize 'result' at the point of declaration</i>	ActiveRecord/ Compiler/src/ Parser.cpp, CppParser/src/ Utility.cpp	2
MISRA-007	WARNING	Variable 'pDeveloper' of type 'int' declared without an initializer <i>Initialize 'pDeveloper' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	2

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'pSeniorDeveloper' of type 'int' declared without an initializer <i>Initialize 'pSeniorDeveloper' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	2
MISRA-007	WARNING	Variable 'pManager' of type 'int' declared without an initializer <i>Initialize 'pManager' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	2
MISRA-007	WARNING	Variable 'fmt' of type 'int' declared without an initializer <i>Initialize 'fmt' at the point of declaration</i>	Benchmark/src/ LoggerBench.cpp	2
MISRA-007	WARNING	Variable 'hasData' of type 'int' declared without an initializer <i>Initialize 'hasData' at the point of declaration</i>	Benchmark/src/ NotificationQueueBench.cpp	2
MISRA-007	WARNING	Variable 'simpleTypes' of type 'const int' declared without an initializer <i>Initialize 'simpleTypes' at the point of declaration</i>	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	1
MISRA-007	WARNING	Variable 'fullName' of type 'std::string' declared without an initializer <i>Initialize 'fullName' at the point of declaration</i>	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	1
MISRA-007	WARNING	Variable 'ns' of type 'auto' declared without an initializer <i>Initialize 'ns' at the point of declaration</i>	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	1

Complexity (4 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
CPP-CX-002	WARNING	Function 'writeInsert' is 83 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	ActiveRecord/ Compiler/src/ ImplGenerator.cpp	1
CPP-CX-002	WARNING	Function 'printTimeSummary' is 110 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	Benchmark/src/ BenchmarkApp.cpp	1
CPP-CX-002	WARNING	Function 'main' is 94 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	Benchmark/src/ BenchmarkApp.cpp	1
CPP-CX-003	WARNING	Function 'printTimeSummary' has 6 parameters (threshold: warning>5, error>8) <i>Introduce a parameter struct or builder pattern to reduce parameter count</i>	Benchmark/src/ BenchmarkApp.cpp	1

Memory Safety (1 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
CPP-MEM-001	WARNING	Raw 'new' expression detected; prefer std::make_unique or std::make_shared <i>Replace 'new T(...)' with 'std::make_unique<T>(...)'</i>	ActiveRecord/ include/Poco/ ActiveRecord/ Query.h, ApacheConnector/ samples/ FormServer/src/ FormServer.cpp, ApacheConnector/ samples/ TimeServer/src/ TimeServer.cpp	4

5. Knowledge Silo Alerts

Files where only one contributor has committed in the last 12 months. These represent bus-factor risk.

No knowledge silos detected.

6. Bug Prediction — Top 10 Highest-Risk Files

XGBoost model bug probability. Files above 0.7 should be prioritised for review.

#	File	Bug Probability	Risk Level	Top Factor
1	Crypto/src/EVPCipherImpl.cpp	99.9%	CRITICAL	misra_findings
2	Crypto/src/RSACipherImpl.cpp	99.9%	CRITICAL	misra_findings
3	Crypto/testsuite/src/EVPTest.cpp	99.9%	CRITICAL	misra_findings
4	Crypto/testsuite/src/RSATest.cpp	99.9%	CRITICAL	misra_findings
5	DNSSD/samples/HTTPTimeServer/src/HTTPTimeServer.cpp	99.9%	CRITICAL	misra_findings
6	Foundation/include/Poco/AbstractCache.h	99.9%	CRITICAL	misra_findings
7	Foundation/include/Poco/Buffer.h	99.9%	CRITICAL	misra_findings
8	Foundation/include/Poco/MPSCQueue.h	99.9%	CRITICAL	misra_findings
9	Foundation/testsuite/src/AnyTest.cpp	99.9%	CRITICAL	misra_findings
10	Foundation/testsuite/src/AutoPtrTest.cpp	99.9%	CRITICAL	misra_findings

7. Refactoring Roadmap

Prioritised by impact score. Complete high-impact, low-effort items first.

Priority	File	Action	Category	Est. Hours	Impact
1	ActiveRecord/Compiler/src/CodeGenerator.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	12.0h	16.0
2	ActiveRecord/Compiler/src/Compiler.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	8.0h	16.0
3	ActiveRecord/Compiler/src/HeaderGenerator.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	6.0h	16.0
4	ActiveRecord/Compiler/src/ImplGenerator.cpp	Reduce cyclomatic complexity by extracting methods and simplifying control flow	COMPLEXITY	3.0h	16.0
5	ActiveRecord/Compiler/src/ImplGenerator.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
6	ActiveRecord/Compiler/src/Parser.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	10.0h	16.0
7	ActiveRecord/include/Poco/ActiveRecord/Query.h	Fix memory safety issues: replace raw pointers with smart pointers and add bounds checks	MEMORY SAFETY	8.0h	16.0
8	ActiveRecord/include/Poco/ActiveRecord/Query.h	Address MISRA C++ compliance violations to meet safety standards	MISRA	6.0h	16.0

Priority	File	Action	Category	Est. Hours	Impact
9	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	58.0h	16.0
10	ActiveRecord/ testsuite/src/ ActiveRecordTestSuite.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
11	ActiveRecord/ testsuite/src/ WinDriver.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
12	ApacheConnector/samples/ FormServer/ src/ FormServer.cpp	Fix memory safety issues: replace raw pointers with smart pointers and add bounds checks	MEMORY SAFETY	4.0h	16.0
13	ApacheConnector/samples/ FormServer/ src/ FormServer.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	14.0h	16.0
14	ApacheConnector/samples/ TimeServer/ src/ TimeServer.cpp	Fix memory safety issues: replace raw pointers with smart pointers and add bounds checks	MEMORY SAFETY	4.0h	16.0
15	ApacheConnector/samples/ TimeServer/ src/ TimeServer.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
16	ApacheConnector/src/ ApacheConnector.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	12.0h	16.0
17	Benchmark/src/ BenchmarkApp.cpp	Reduce cyclomatic complexity by extracting methods and simplifying control flow	COMPLEXITY	9.0h	16.0

Priority	File	Action	Category	Est. Hours	Impact
18	Benchmark/src/BenchmarkApp.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	26.0h	16.0
19	Benchmark/src/LoggerBench.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	36.0h	16.0
20	DNSSD/Avahi/src/AvahiBrowserImpl.cpp	Reduce cyclomatic complexity by extracting methods and simplifying control flow	COMPLEXITY	21.0h	12.0

Generated by cppulse — C++ Codebase Health Analyzer